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HONOURS PROJECT

Mean-Variance Hedging With Neural Network Using
Heston Model

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Statistical Sciences

Declaration of Authorship

We, Masande Besi and Lwando Mbanguta, declare that this project titled, "Mean-Variance Hedging with Neural Network using Heston Model" and the work presented in it are our own work. We confirm that:

- This work was done wholly and mainly while in candidature for BBus Sci in Ana degree at the Univeristy of Cape Town.
- Where we have consulted the published work of others, this is always clearly attributed and we have acknowledged all main sources of help.
- Where we have used the words of others, we have indicated this by the use of quotation marks. With the exception of such quotations, this project is entirely our own work.

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Abstract

We investigate the effectiveness of hedging a European basket swap option using the Heston model within a contractual framework involving two parties. The contractual agreement stipulates that at a time T , if the value of the stock owned by party one exceeds that of party two, a trade occurs otherwise the option is not exercised. We make use of the Heston Model as an underlying sample path generator to optimise the hedging strategy, considering factors such as the stochastic volatility of each stock, the co-variation between the stocks and their volatilities, and the correlation between the two stocks.

Specifically, we explore how covariance between the two underlying stocks, as well as their respective covariances with their volatilities influence option pricing and hedging outcomes. To estimate the parameters of the multivariate Heston model, our first step is to discretise the model utilising Euler's discretisation method. Thereafter, employ Maximum Likelihood Estimation on the joint multivariate Normal distribution of the two stocks and the volatilities to estimate the parameters of the model.

Lastly, we employ neural networks to hedge the option and compare the results with a hedge based on the Black-Scholes model.

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1 Introduction

In financial risk management, hedging plays an important role in mitigating the uncertainties associated with asset price fluctuations. Traditional hedging techniques, such as the Black-Scholes model, have laid the foundation for modern financial theory, yet they often fall short in capturing the complex dynamics of real-world markets. One of the primary limitations of these classical models is their assumption of constant volatility, which contradicts empirical observations of market behaviour. To address this, the Heston model, introduced by Steven Heston in 1993, incorporates stochastic volatility, providing a more realistic representation of asset price movements.

Even with the Heston model, finding the best hedging strategy is still difficult. Mean-variance hedging, which aims to minimise the squared error in hedging, is a common and reliable method. But it traditionally uses linear techniques, which may not fully capture the complex relationships in financial data. This is where new developments, like neural networks, come in. Neural networks are good at recognising complex and non-linear patterns, making them a great tool for modeling the complexities of financial markets.

This research aims to explore the integration of neural networks into mean-variance hedging frameworks, using the Heston model as an underlying sample path generator. The Heston model's ability to simulate realistic asset price trajectories provides a robust testing ground for evaluating the performance of neural network-based hedging strategies. The primary objective is to determine whether neural networks can outperform traditional methods in minimising hedging errors and managing financial risk.

To achieve this, we train neural networks on data produced by the Heston model. These networks are built to learn the best hedging strategies that reduce errors over time. The performance of these strategies is then tested and compared with traditional Black-Scholes hedging methods. The study also considers how neural network-based hedging can be used in real financial markets, looking at issues like speed, ease of understanding, and how well the models can adapt to changing market conditions.

In summary, this research aims to show how neural networks can improve mean-variance hedging strategies. By combining advanced machine learning with traditional financial models, the study hopes to offer a new way to manage financial risk in today's complex and volatile markets.

2 Theoretical Background

For a comprehensive understanding of derivatives, basket swap options, and the models used in this project, we consulted and reference a range of seminal and recent works. Readers interested in foundational principles of derivative pricing can consult the work of [Hull et al. \(2009\)](#) in *Options, Futures, and Other Derivatives*, which provides an extensive overview of derivative markets and option pricing.

In terms of stochastic volatility models, the Heston model has become a widely cited standard. Heston's original paper, *A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options* [Heston \(1993\)](#), offers a thorough explanation of the model and its implications. For a deeper understanding of extensions and applications of the Heston model, the work by [Gatheral \(2011\)](#) is also recommended.

The Black-Scholes model, which provides the foundation of many option pricing theories, was first introduced in *The Pricing of Options and Corporate Liabilities* by [Black and Scholes \(1973\)](#) and further elaborated in [Merton \(1973\)](#), *Theory of Rational Option Pricing*.

The application of neural networks in finance, particularly in option pricing and hedging, has got increasing interest and recent works as *Deep Hedging* by [Buehler et al. \(2019\)](#) and *Machine Learning for Financial Engineering* by [Ruf and Wang \(2019\)](#) explore the use of neural network for dynamic hedging strategies and provide insights into the advantages of machine learning approaches over traditional methods.

Readers are encouraged to explore these references for a detailed understanding of the models and techniques discussed in the paper.

3 Literature Review

The Heston Model, introduced by Steven Heston, has become one of the cornerstones of quantitative finance due to its ability to capture the dynamics of the asset prices and their volatility. It extends the Black-Scholes Model by introducing stochastic volatility, allowing for more accurate modelling of the stock prices, especially in markets with varying levels of volatility.

Numerous studies have applied the Heston Model to price and hedge various types of options, including basket options. For example [Lin et al. \(2018\)](#) investigated the pricing of basket options under the Heston Model and highlighted the importance of accurately estimating model parameters for effective hedging strategies. Their finding emphasised the need for reliable parameter estimation techniques to improve performance.

Hedging basket options requires sophisticated strategies that account for the co-variation between the underlying stocks and their respective volatilities. The Heston Model offers a flexible framework for designing such strategies. [Xiaoshan \(2020\)](#) examined the effectiveness of the different hedging strategies for basket options using the Heston Model and found that delta hedging, combined with variance swaps, can effectively reduce risk exposure and improve performance.

In the real-world financial markets, contractual agreements often govern the trading and hedging activities of multiple parties. The effectiveness of the hedging strategies can be influenced by the terms of these agreements. [He and Zhu \(2019\)](#) analysed the impact of contractual constraints on the hedging strategies for basket options using the Heston Model. Their study highlighted the importance of aligning hedging strategies with the contractual obligations to maximise hedging effectiveness and minimise associated costs.

Accurate parameter estimation is important for the effective implementation of the Heston Model in option pricing and hedging. Various techniques, such as Maximum Likelihood Estimation (MLE and Monte Carlo simulations, have been employed to calibrate model parameters based on market data. [Chen et al. \(2019\)](#) proposed a novel approach by combining MLE with particle swap optimisation to improve parameter estimation accuracy for the Heston Model, this improved the effectiveness of the hedging strategies for basket options.

While significant progress has been made in understanding the dynamics of option pricing and hedging under the Heston Model, challenges remain in accurately estimating model parameters and designing effective hedging strategies that align with contractual obligations. Further research in these areas is necessary to improve the practical applicability of the Heston Model in real-world financial markets.

4 Methodology

4.1 Heston Model

The formulas proposed by Steven Heston to describe the movement of the asset prices, where both the asset price and its volatility follow random Brownian motion process.

$$dS_t = \mu S_t dt + \sqrt{V_t} S_t dW_t^S \quad (1)$$

$$dV_t = \kappa(\theta - V_t)dt + \xi\sqrt{V_t}dW_t^V \quad (2)$$

$$d\langle W_t^S, W_t^V \rangle = \rho dt \quad (3)$$

where:

- S_t : the asset price at time t .
- V_t : volatility of the stock price (instantaneous variance).
- μ : risk-free interest rate.
- ξ : variance/volatility of the volatility V_t (non-negativity).
- θ : long-term price variance.
- κ : rate of reversion to the long-term price variance i.e. the rate at which V_t reverts to θ (non-negative).
- dt : indefinitely small time increment.
- W_t^S and W_t^V : Brownian motion for the stock price and its volatility respectively.
- ρ : the correlation between the Brownian motions, and $\rho \in (-1, 1)$.

By making use of Euler discretisation, we essentially approximate the continuous-time stochastic process of the asset prices and the volatilities into a discrete time framework, this makes it feasible to do analysis using the Heston Model. More importantly, the Heston Model treats the movement of the asset prices as the continuous time process while they occur in discrete time, which is another motivation to discretise the asset price movement.

Moreover, there are two conditions that should be satisfied while calibrating the model.

- The Feller condition: $2\kappa\theta > \xi^2$, this ensures that the instantaneous variance, V_t , is strictly positive.

- The covariance matrix, Σ_t , is positive definite. This implies that the eigenvalues λ_i for the covariance matrix are strictly positive and $\det(\Sigma_t) > 0$.

4.2 Likelihood Function

The two stocks and the volatilities for $i \in \{1, 2\}$:

$$dS_t^i = \mu^i S_t^i dt + \sqrt{V_t^i} S_t^i dW_t^{S^i} \quad (4)$$

$$dV_t^i = \kappa^i (\theta^i - V_t^i) dt + \xi^i \sqrt{V_t^i} dW_t^{V^i} \quad (5)$$

The correlations of the stocks with their volatilities and the correlation between the two stocks:

$$d\langle W^{S^1}, W^{S^2} \rangle_t = \rho_{s^1 s^2} dt \quad (6)$$

$$d\langle W^{S^i}, W^{V^i} \rangle_t = \rho_{s^i v^i} dt, \text{ and} \quad (7)$$

$$d\langle W^{S^i}, W^{V^j} \rangle_t = 0 \quad \text{for } i \neq j \quad (8)$$

Discretise the time interval with Δt such that:

$$S_{t+\Delta t}^i = S_t^i + \mu^i S_t^i \Delta t + \sqrt{V_t^i} S_t^i \sqrt{\Delta t} Z_i^S \quad (9)$$

$$V_{t+\Delta t}^i = V_t^i + \kappa^i (\theta^i - V_t^i) \Delta t + \xi^i \sqrt{V_t^i} \sqrt{\Delta t} Z_i^V \quad (10)$$

Now define multivariate random variable $\mathbf{X}_t = (S_t^1, S_t^2, V_t^1, V_t^2)$, such that $\mathbf{X}_{t+\Delta t} | \mathbf{X}_t \sim MVN(\mathbf{m}_t, \Sigma_t)$, where:

$$\mathbf{m}_t = \begin{bmatrix} S_{t-1}^1 + \mu^1 S_{t-1}^1 \Delta t \\ S_{t-1}^2 + \mu^2 S_{t-1}^2 \Delta t \\ V_{t-1}^1 + \kappa^1 (\theta^1 - V_{t-1}^1) \Delta t \\ V_{t-1}^2 + \kappa^2 (\theta^2 - V_{t-1}^2) \Delta t \end{bmatrix}$$

and

$$\Sigma_t = \begin{bmatrix} (\sqrt{V_{t-1}^1} S_{t-1}^1)^2 \Delta t & \sqrt{V_{t-1}^1} \sqrt{V_{t-1}^2} S_{t-1}^1 S_{t-1}^2 \rho_{s^1 s^2} \Delta t & V_{t-1}^1 S_{t-1}^1 \xi^1 \rho_{s^1 v^1} \Delta t & 0 \\ \sqrt{V_{t-1}^1} \sqrt{V_{t-1}^2} S_{t-1}^1 S_{t-1}^2 \rho_{s^1 s^2} \Delta t & (\sqrt{V_{t-1}^2} S_{t-1}^2)^2 \Delta t & 0 & V_{t-1}^2 S_{t-1}^2 \xi^2 \rho_{s^2 v^2} \Delta t \\ V_{t-1}^1 S_{t-1}^1 \xi^1 \rho_{s^1 v^1} \Delta t & 0 & (\xi^1 \sqrt{V_{t-1}^1})^2 \Delta t & 0 \\ 0 & V_{t-1}^2 S_{t-1}^2 \xi^2 \rho_{s^2 v^2} \Delta t & 0 & (\xi^2 \sqrt{V_{t-1}^2})^2 \Delta t \end{bmatrix}$$

Thus the joint probability density function of $\mathbf{X}_{t+\Delta t} | \mathbf{X}_t$ is therefore

$$f(\mathbf{x}_{t+\Delta t} | \mathbf{x}_t) = \frac{1}{(2\pi)^{\frac{4}{2}} |\Sigma_t|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t)^T \Sigma_t^{-1} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t) \right] \quad (11)$$

The likelihood is then defined as the product of the probability density functions of each observation of the random variable. Let $\boldsymbol{\eta} = [\mu^i, k^i, \theta^i, \xi^i, \rho_{s^i s^j}, \rho_{s^i v^i}]$, encapsulating all the parameters.

$$\begin{aligned} L(\boldsymbol{\eta}) &= \prod_{t=1}^n f(\mathbf{x}_{t+\Delta t} | \mathbf{x}_t) \\ &= \prod_{t=1}^n (2\pi)^{-2} |\boldsymbol{\Sigma}_t|^{-\frac{1}{2}} \exp \left[-\frac{1}{2} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t)^T \boldsymbol{\Sigma}_t^{-1} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t) \right] \end{aligned}$$

The log-likelihood function

$$\log[L(\boldsymbol{\eta})] = -2n \log(2\pi) - \frac{1}{2} \sum_{t=1}^n \log |\boldsymbol{\Sigma}_t| - \frac{1}{2} \sum_{t=1}^n (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t)^T \boldsymbol{\Sigma}_t^{-1} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t) \quad (12)$$

4.3 Data Generation

The idea of data generation is, we randomly generate the stock prices and volatilities of the two correlated assets over a one-year horizon with daily steps, which we assume is observed stock prices prior to the contractual agreement. The initial parameters were: risk-free interest rate $\boldsymbol{\mu} = [0.05, 0.05]$, rate of reversion $\boldsymbol{\kappa} = [2, 2]$, long-term variance $\boldsymbol{\theta} = [0.015, 0.05]$, volatility of the volatility $\boldsymbol{\xi} = [0.1, 0.02]$. The correlation were $\rho_{s^1 v^1} = 0.3$, $\rho_{s^2 v^2} = 0.3$ and $\rho_{s^1 s^2} = -0.2$. The

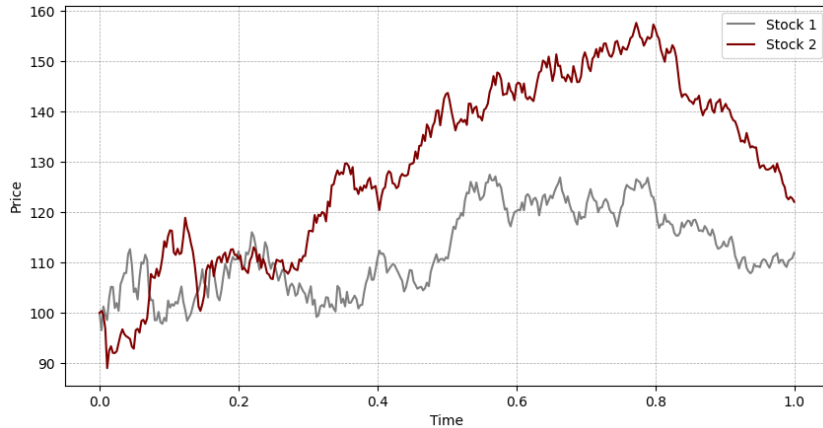


Figure 1: Simulated stock prices under the initial parameters

stock prices and volatilities were initialised at 100 and 0.2, respectively. Figure 1 depicts the simulated stock price trajectories, and the complete code is available in the appendix. With this information of the asset performances, the parties would keenly be interested in understanding the future price paths of the respective stocks, as they impact the payoff structure and the risk profile of the option. Investors can therefore make informed decisions on whether to enter into,

hold, or adjust their positions in the basket option, thereby optimising their financial strategies and outcomes.

Prior to entering into the basket swap contract, understanding the potential future stock price paths may prove significant for several reasons. One may aim to hedge against adverse movements in the underlying stock prices and by simulating possible future stock prices, one can gain insights into potential scenarios where the stocks may underperform or outperform, allowing them to adjust their risk exposure accordingly. Additionally, a party may have specific

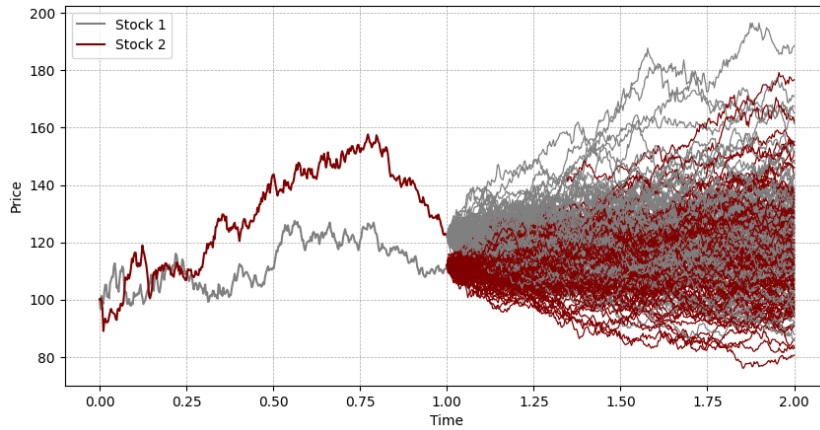


Figure 2: Simulated trajectories of stock prices after $T = 1$, showcasing the range of potential outcomes for basket options.

investment goals or obligations tied to the performance of the underlying assets. By analysing the simulated price paths, they can assess the likelihood of meeting these objectives and make informed decisions regarding the basket swap option.

From the perspective of party two entering into the basket swap option, the trading strategy $\Pi = (S_T^1 - S_T^2)^+$ ensures that party two takes a long position when the value of the party one's stock is higher than the value of their stock, S_t^2 , and a short position otherwise. Given the simulated stock prices in Figure 2, party two can analyse the different scenarios using the trading strategy at which they gain a positive payoff or do not enter into the contract. To visualise these scenarios, the payoff distribution using the simulated future stock prices and the payoff surface are illustrated in Figure 3. The payoff surface shows the different combinations of the respective stock prices whereby party two enters into the swap option and observe a long position and the scenarios where they do not enter into the swap option. This allows party two to evaluate the potential profitability of entering into the swap option across a wide range of price scenarios. The histogram, shows the distribution of payoffs using the given trading strategy, and the frequency of payoffs if party two enters into the basket swap option. By visualising the frequency and magnitude of payoffs, party two can assess the risk-return profile associated with the option across a range of scenarios.

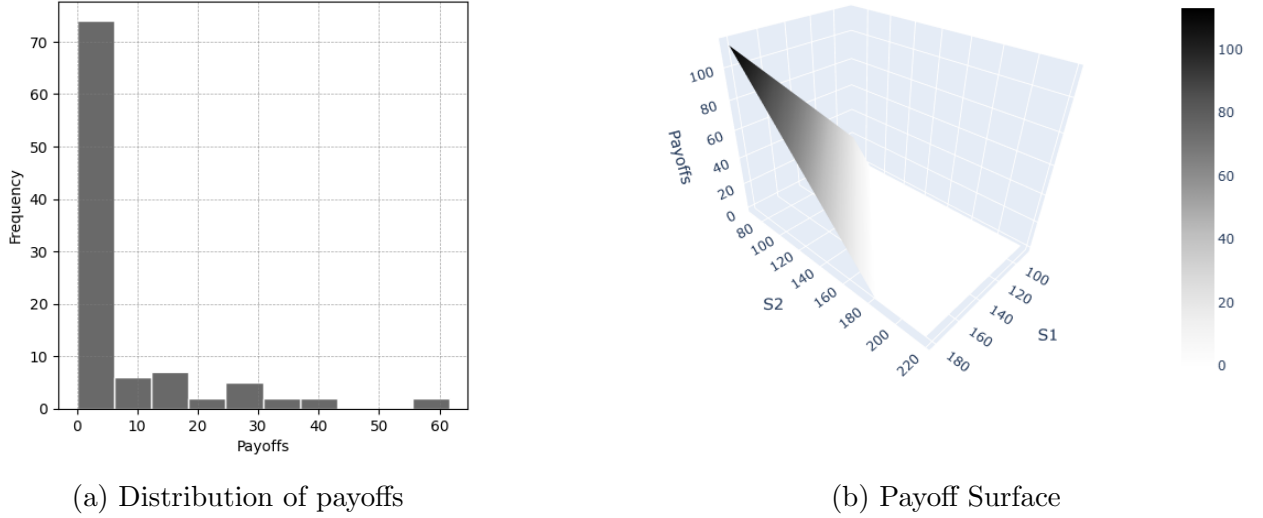


Figure 3: The left sub-figure displays the distribution of payoffs, while the right sub-figure illustrates the payoff surface. These visualisations provide valuable insights for Party two entering into the basket swap contract, aiding in risk assessment.

4.4 Parameter Estimation

In this short section, we explain how to estimate the parameters of the multivariate Heston model from a data set of asset prices. A lot of ground has been covered in the Likelihood Function section. Specifically, we have defined the log likelihood function, which incorporates the key parameters of the model, and we are interested in estimating eleven of them. The parameters of interest;

$$[\mu^1, \mu^2, \kappa^1, \kappa^2, \xi^1, \xi^2, \theta^1, \theta^2, \rho_{s^1 s^2}, \rho_{s^1 v^1}, \rho_{s^2 v^2}] \quad (13)$$

These parameters are estimated by maximising the log-likelihood function given the observed data. The estimation process involves initialising the parameters with plausible values and then using numerical optimisation techniques. The constraints of the parameters ensure that the estimated values adhere to theoretical and practical bounds. The Feller condition ensures the positivity of the variance process, and the determinant of the covariance matrix is maintained within specified bounds to ensure it remains positive semi-definite. This estimation process allows for accurate modeling of asset prices and their volatilities, and enabling the application of the model in various financial analyses and derivative pricing tasks.

5 Hedging: Neural Networks vs Black-Scholes

Neural networks are well-known for their strong ability to predict outcomes and are being used more in finance to find non-linear patterns and give useful insights from data. They are composed of layers of interconnected nodes that process and transform input data to produce predictions. Their flexibility and ability to generalise complex patterns make them effective in dealing with the inherent uncertainties of financial markets.

The Universal Approximation Theorem for neural networks asserts that a feed-forward neural network with a single hidden layer containing a sufficient number of nodes can approximate any continuous function to a desired degree of accuracy, given appropriate activation functions. This powerful property explains their applicability to a wide range of financial problems, where accurate modeling of complex patterns is important.

Neural networks have shown promise in hedging strategies, such as in the work by [Buehler et al. \(2019\)](#), where they outperform traditional methods by offering a more scalable and adaptive approach, especially in capturing complex market dynamics that standard models struggle with. This flexibility and ability to handle large datasets make neural networks a valuable tool in financial decision-making.

5.1 Amazon and Google Historical Data

For this project, we have chosen data from Amazon and Google and we assume that party one is holding Google stock and is entering into a basket swap option that includes both Google and Amazon stocks. These two stocks have been heavily traded in the past and their market data is readily available, making them ideal for our analysis.

One key reason for choosing these stocks is that the volatility of both stocks is clearly not constant over time. The stock prices have shown significant fluctuations, especially during periods of market instability or company-specific events as seen in Figure 4. Due to this, the standard Black-Scholes model, which assumes constant volatility, may not provide an accurate representation of the stocks.

The Heston model, on the other hand, is well-suited for this type of situation because it allows for changing volatility over time. This flexibility makes it an appropriate model for handling varying levels of uncertainty associated with these stocks.



Figure 4: Amazon and Google historical stock prices [Bloomberg LP \(2024\)](#).

The stochastic volatility for both stocks was obtained by first calculating the daily log returns r_t of the stock prices S_t .

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right)$$

This gives the natural logarithm of the ratio of stock prices from one day to the next. After obtaining the returns, we compute the rolling volatility over a 14-day window using the standard deviation σ_t of the log returns:

$$\sigma_t = \text{std}(r_{t-14:t}) \cdot \sqrt{365}$$

Here $\text{std}(r_{t-14:t})$ is the standard deviation of returns over the past 14 days, and we multiple by $\sqrt{365}$ to annualise the volatility, converting it into a yearly measure. The rolling volatility gives us a time-varying estimate of how volatile the stock price is over time. The importance of this stochastic volatility lies in capturing the nature of the risk associated with stock prices. Unlike models that assume constant volatility, the Heston model assumes that volatility follows a stochastic process as seen in equation 2. This framework allows us to account for time-varying volatility, where the volatility itself is subject to randomness. This feature makes the Heston model better suited for real-world financial data. This historic stochastic volatility for both stocks is displayed in Figure 5.

By leveraging both the observed stock prices and their corresponding volatilities, we apply Maximum Likelihood Estimation (MLE), as described in the Parameter Estimation section, to obtain the multivariate Heston model parameters. MLE is used to estimate the parameters by fitting the model to the historical price and volatility data, maximising the likelihood of observing the given data under the specified model.

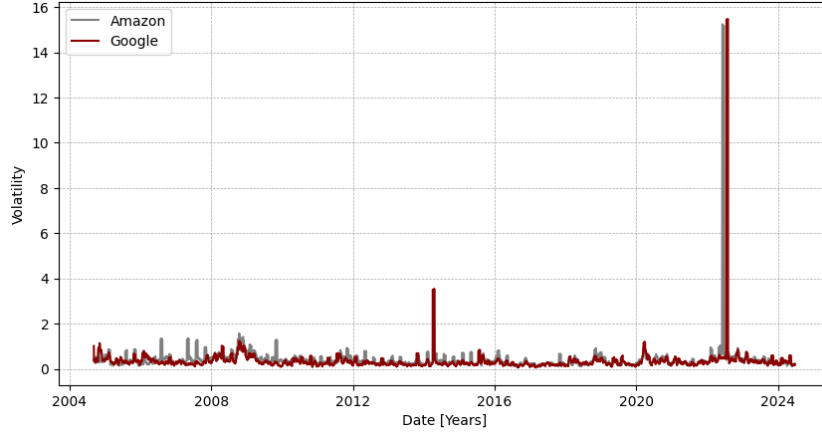


Figure 5: Historic stochastic volatility of Amazon and Google stocks

The estimated parameters for the multivariate Heston model are:

$$[\mu^1, \mu^2, \kappa^1, \kappa^2, \xi^1, \xi^2, \theta^1, \theta^2, \rho_{s^1 s^2}, \rho_{s^1 v^1}, \rho_{s^2 v^2}]$$

The estimated values (from the code in Appendix A.2) are:

$$[0.194, 0.268, 4.597, 2.041, 0.179, 0.493, 0.772, 0.407, 0.848, -0.010, -0.012]$$

These values capture the stochastic nature of both stock prices and their volatilities under the Heston framework.

5.2 Stock Price Simulations

For the basket swap option, we assume we are at the end of May, with time to maturity of the option being the end of June. The holder of Google stock, entering into the option seeks to understand how the two stocks might behave over the month of June. Under the parameters estimated in the previous section using the multivariate Heston model, we simulate many possible future stock price paths across this period.

These simulations are important because they capture a wide range of possible outcomes for stock prices under the stochastic volatility model, accounting for the randomness inherent in both price and volatility. By simulating these paths, the party entering into the contract can observe the potential movements of these stocks and thus, make informed decisions about risk management.

Figure 6 illustrates only 100 of these simulated price paths, out of many more that were generated. Each represents a possible scenario for the two stocks over the month of June. The party entering into this contract would like to hedge against all of these scenarios to mitigate the risk associated with unfavourable movements in the stock prices.

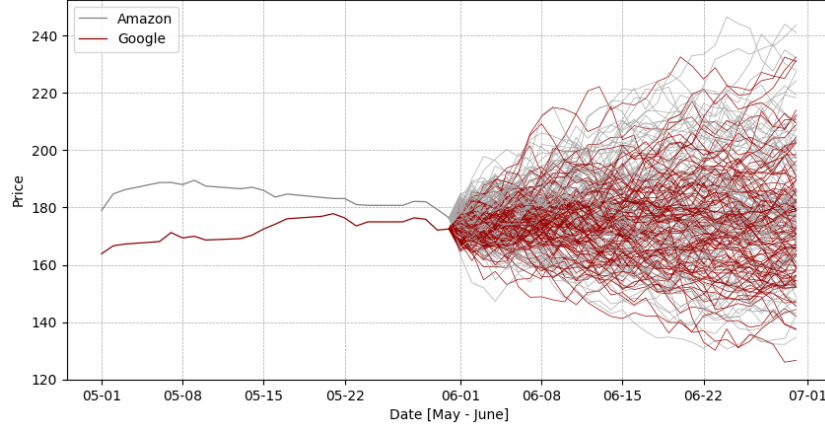


Figure 6: May historic prices and June simulated stock prices under Heston model.

5.3 Option Pricing with Black-Scholes

The goal of this section is to explain how the Black-Scholes model is used to price a basket swap option on two underlying assets—Amazon and Google, this follows from [Mavuso \(2019\)](#). In a basket swap option, the payoff is based on the difference in performance between the two assets, $\Pi = (S_T^1 - S_T^2)^+$. The holder of the option receives a payoff if the value of one asset exceeds the other, based on the final value at maturity.

The Black-Scholes model provides a framework for pricing options based on the assumption of constant volatility and log-normal distribution of stock prices. While this assumption is not realistic for real-world assets due to stochastic volatility, the Black-Scholes formula remains a useful tool for understanding and benchmarking option prices.

For pricing the basket swap option, we use a modified version of the Black-Scholes formula that accounts for the correlation between the two underlying assets. The key inputs are the stock prices, volatilities of each stock, time to maturity, and the correlation between the stocks.

The two critical factors for pricing are the quantities d_1 and d_2 , which are defined as:

$$d_1 = \frac{\ln\left(\frac{S_t^1}{S_t^2} + \frac{1}{2}\sigma^2 T\right)}{\sigma\sqrt{T}} \quad (14)$$

$$d_2 = d_1 - \sigma\sqrt{T} \quad (15)$$

Where:

- S_t^1 and S_t^2 are the stock prices at time t
- σ is the combined volatility of the two assets, adjusted by the correlation ρ_{s^1, s^2}

- T is time to maturity

In this case, the combined volatility σ is derived using the formula:

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho_{s^1s^2}\sigma_1\sigma_2} \quad (16)$$

Where σ_1 and σ_2 are the volatilities of each stock, and $\rho_{s^1s^2}$ is the correlation between the two stocks. The price, H_t , of the basket swap option is then given by the Black-Scholes formula:

$$H_t = S_t^1 \cdot \Phi(d_1) + S_t^2 \cdot \Phi(d_2) \quad (17)$$

Where Φ is the cumulative distribution function of the standard normal distribution. This formula represents the expected value of the different between the two stock prices, weighted by the likelihood of different outcomes.

5.4 Black-Scholes Hedge

In addition to pricing the option, the Black-Scholes model also provides the hedge ratios for each stock. The hedge ratios, represented as $\boldsymbol{\varphi} = (\varphi^1, \varphi^2)$, define the trading strategy and derived from the partial derivatives of the option price with respect to each stock price:

$$\varphi^1 = \Phi(d_1) \quad (18)$$

$$\varphi^2 = -\Phi(d_2) \quad (19)$$

These hedge ratios indicate the number of shares of each stock that need to be held in the hedging portfolio to minimise risk. These ratios can be calculated across the month of June to obtain how many shares of Amazon to hold, φ_t^1 , and how many shares of Google to hold, φ_t^2 , at time t .

The value of the trading strategy is a stochastic process $V(\boldsymbol{\varphi}) = \{V(\varphi_t^1, \varphi_t^2) : t \in [0, T]\}$ defined by for $i \in \{1, 2\}$:

$$V_t(\boldsymbol{\varphi}) := \sum_{i=1}^{d=2} \varphi_t^i S_t^i \quad t \geq 1 \quad (20)$$

This equation represents the dot product of the strategy and the change in asset prices at time t . This trading strategy is used to hedge the basket swap option, where the objective is to track the option's price over time. By holding a certain number of shares of each stock φ_t^1 and φ_t^2 , we adjust the portfolio such that its value aligns with the changing option price.

In our case, we assume that the hedge portfolio is a self-financing trading strategy such that the only changes to the portfolio's value come from the gains process, there are not external cash flows into or out of the portfolio. The gains process, denoted by $G(\boldsymbol{\varphi})$, captures the cumulative gains from the trading strategy:

$$G_0(\boldsymbol{\varphi}) := 0, \quad G_t(\boldsymbol{\varphi}) := \sum_{k=1}^t \varphi_k \cdot \Delta S_k \quad \text{for } k = 1, 2, \dots, T. \quad (21)$$

This represents the changes in the portfolio value due to price movements of the assets over time. Thus the value of the trading strategy

$$V_t(\boldsymbol{\varphi}) = V_0(\boldsymbol{\varphi}) + G_t(\boldsymbol{\varphi}) \quad \text{for } t = 0, 1, 2, \dots, T. \quad (22)$$

This condition ensures that the portfolio evolves solely based on asset price changes and trading decisions, where $V_0(\boldsymbol{\varphi}) = S_0^1 \cdot \Phi(d_1) + S_0^2 \cdot \Phi(d_2)$, is the option price at the beginning ($t = 0$). The holding are used to continuously adjust the portfolio, ensuring that the value of the trading strategy $V_t(\boldsymbol{\varphi})$ tracks the option price at every point in time. The gains process reflects the cumulative adjustments made through the hedging strategy to offset risks from stock price fluctuations.

The profit and loss (PNL) at the end of the hedging period is the difference between the portfolio value and the option payoff at maturity, ensuring that the trading strategy is designed to closely track the price of the option over its life.

$$\text{PNL} = V_T(\boldsymbol{\varphi}) - H_T \quad (23)$$

The objective is to obtain a distribution of PNL from the many path simulations, and for mean-variance hedging, we would like for this distribution to be centered around zero with minimal variance.

5.5 Neural Network Hedge

Another way to obtain the hedge ratios is through a neural network. For each path simulation, we use S_t^1, S_t^2 and τ (time to maturity of the option) as inputs to the network to predict the holdings:

$$f^{NN}(S_t^1, S_t^2, \tau) = (\varphi_t^1, \varphi_t^2) \quad (24)$$

The value of the self-financing portfolio is still obtained using the previously equations (20) and (22) and the option price, H_t , is obtained using Black-Scholes formula (17). The neural network predicts the optimal number of units of each stock to hold at time t in order to track the option price. The aim is to minimise the difference between the predicted change in portfolio value, based on the trading strategy, and the actual change in the option price.

$$\text{Mean Squared Error: } \mathbb{E}[(\Delta V_t - \Delta H_t)^2] \quad (25)$$

The goal is to minimise this loss such that the neural network learns to adjust $(\varphi_t^1, \varphi_t^2)$ to effectively hedge the option i.e. the value of the trading strategy at time t , V_t , tracks the option price by matching the change in option price (ΔH_t) at time t .

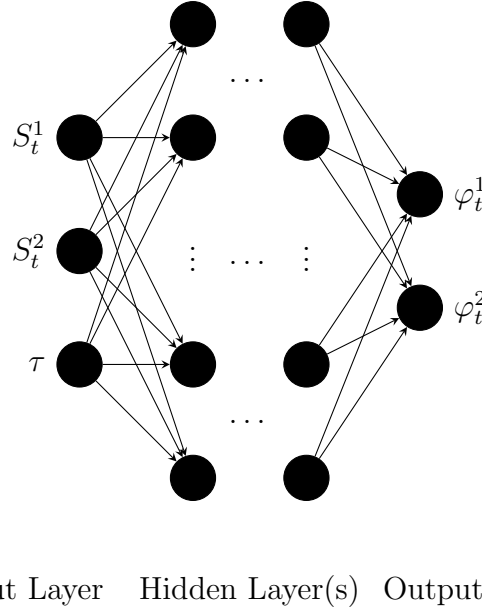


Figure 7: A generalized standard feed-forward neural network structure for this problem.

Figure 7 illustrates a generalised structure for a standard feed-forward neural network excluding the bias terms. For neural network to predict the optimal holdings for each stock and minimise the loss function we perform hyper-parameter tuning for following;

- Number of hidden layers for the neural network.
- Number of nodes in each hidden layer.
- Activation functions to be used on each hidden layer.
- Optimiser for backpropagation, focusing on Stochastic Gradient Descent (SGD) and Adaptive Moment Estimation (Adam).
- Regularisation methods, choosing between ridge or lasso.

While here are many other hyperparameters that can be tuned, such as the learning rate for backpropagation, we use a scheduler in PyTorch that adjusts the learning process. The output layer makes use of hyperbolic tangent function to allow for short-selling such that $\varphi_t^i \in [-1, 1]$ for each stock $i \in \{1, 2\}$.

As [Robbertze \(2021\)](#) put it, in machine learning it is standard procedure to split the dataset into a training and validation sets which one trains a model on the training data to evaluate in-sample performance, while the validation dataset is used to ass the model's out-of-sample performance. This approach helps obtain a model that performs well on unseen data and can be used for predictions. Due to lack of data constraints that we are faced with in the financial

industry we simulate data for training and likewise simulate data to be used for validation.

Ultimately, our objective is to obtain a distribution of PNL (23) and compare this distribution to that obtained with Black-Scholes model in the previous section. We aim for a PNL distribution centered around zero with minimal variance, thereby achieving mean-variance hedging. The Black-Scholes hedge relies on a well-established formula that is deterministic and based on the assumption of constant volatility and normally distributed returns. On the other hand, the neural network approach is more flexible, allowing for the learning of complex, non-linear relationships in the data. This can potentially lead to better hedging performance under more realistic market conditions.

6 Results and Analysis

This section, presents the results of our analysis comparing the hedging performance of the Black-Scholes model and the neural network approach for a basket swap option involving Amazon and Google stocks. We aim to assess the profit and loss (PNL) distributions generated by each approach, focusing on their effectiveness in minimising risk.

6.1 Neural Network Training

To train a neural network to hedge, data preparation involved simulating training, validation and finally test dataset for comparison with Black-Scholes hedge. Each path simulation contained a time series data for the two assets S_t^1 and S_t^2 , the time to maturity of the option τ , and the corresponding option prices H_t . The training and validation datasets were generated with 1000 simulated paths, while 10000 paths were reserved for testing.

The architecture of the neural network consisted of five fully connected hidden layers. Each hidden layer used different activation functions based on optimal selections from hyperparameter tuning, while the output layer used hyperbolic tangent function to allow for the possibility of short-selling. A custom loss function was developed to account for the daily PNL based on changes in stock prices, ΔS_t^i , and the change in option price ΔH_t as seen in (25). Additionally, regularisation terms were incorporated to control model complexity and prevent overfitting.

To ensure optimal model performance, we conducted hyperparameter tuning using the Optuna framework. The objective was to minimise the total loss by selecting the best combination of the following parameters: hidden layer sizes, activation functions for each layer, and the type of regularisation (L_1 or L_2). Optuna was used to conduct 10 trials, and the best-performing hyperparameters were identified based on the lowest loss value. These hyperparameters formed the final model, which was subsequently trained on the training dataset and validated using the validation dataset to approximate out-of-sample performance. The hyperparameters selected for the final model included:

- Hidden layer sizes: [240, 220, 180, 160]
- Activation functions: [ReLU, Sigmoid, Tanh, ReLU]
- Optimiser: Adam
- Regularisation and regularisation value: L_2 and $\lambda = 0.01$

The final model was trained over 10000 epochs, during which the optimiser, Adam, adjusted weights to minimise the loss function, and a learning rate scheduler reduced the learning rate when performance plateaued, further enhancing convergence. During the training process, the model was evaluated on the average loss on the validation data, providing insight into its ability to generalise to unseen data.

6.2 Profit and Loss Distribution Comparison

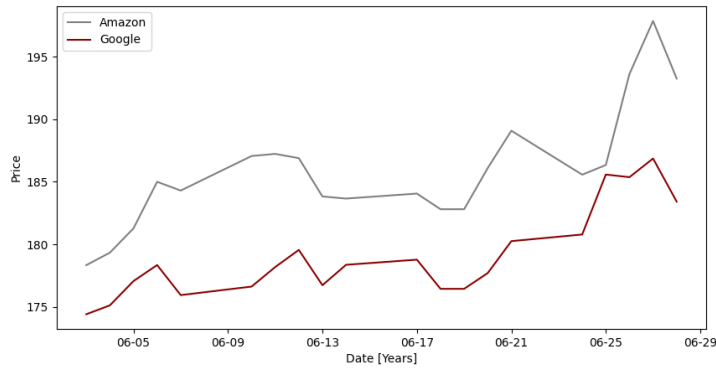


Figure 8: Actual observed June stock prices

7 Conclusion

7.1 Summary of Findings

7.2 Limitations and Future Research Directions

References

- Black, F. and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of political economy*, 81(3):637–654.
- Bloomberg LP (2024). End of day stock prices of amazon and google from 18 august 2004 to 25 june 2024. Bloomberg Terminal. Retrieved 25 June, 2024 from UCT Bloomberg terminal.
- Buehler, H., Gonon, L., Teichmann, J., and Wood, B. (2019). Deep hedging. *Quantitative Finance*, 19(8):1271–1291.
- Chen, Y., Fan, J., Ma, C., and Wang, K. (2019). Spectral method and regularized mle are both optimal for top-k ranking. *Annals of statistics*, 47(4):2204.
- Gatheral, J. (2011). *The volatility surface: A Practitioner’s Guide*. John Wiley and Sons, Inc.
- He, X.-J. and Zhu, S.-P. (2019). Variance and volatility swaps under a two-factor stochastic volatility model with regime switching. *International Journal of Theoretical and Applied Finance*, 22(04):1950009.
- Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The review of financial studies*, 6(2):327–343.
- Hull, J. et al. (2009). *Options, futures and other derivatives/John C. Hull*. Upper Saddle River, NJ: Prentice Hall,.
- Lin, L., Li, Y., and Wu, J. (2018). The pricing of european options on two underlying assets with delays. *Physica A: Statistical Mechanics and its Applications*, 495:143–151.
- Mavuso, M. (2019). *Mathematical Finance*. University of Cape Town, Cape Town, South Africa.
- Merton, R. C. (1973). Theory of rational option pricing. *The Bell Journal of economics and management science*, pages 141–183.
- Robbertze, Y. (2021). Neural network libor market model for pricing and hedging interest rate derivatives. Department of Statistical Sciences.
- Ruf, J. and Wang, W. (2019). Neural networks for option pricing and hedging: a literature review. *arXiv preprint arXiv:1911.05620*.
- Xiaoshan, H. (2020). Interpretability in deep learning for heston smile modeling calibration.